Retail Demand Forecasting System with Time-Series Models and FastAPI Deployment

Project report | Muhammad Faizan Farooq

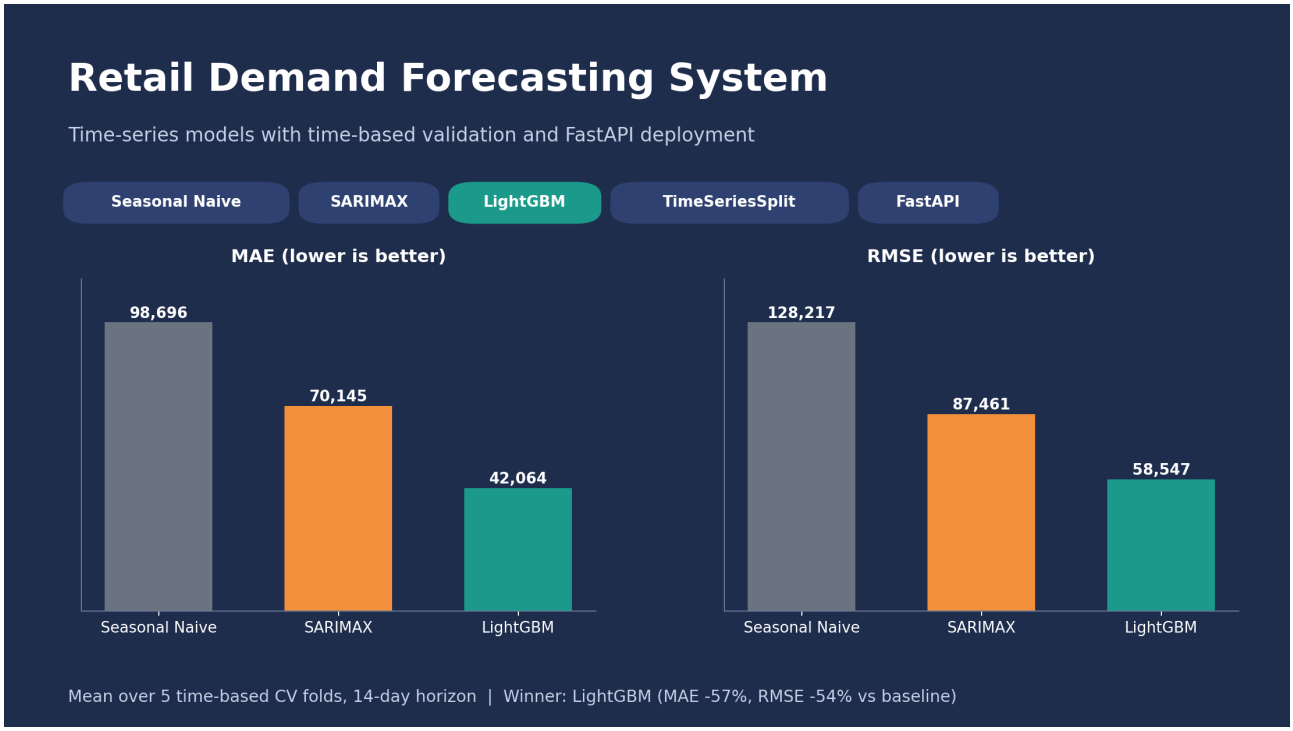


Figure 1. Project overview: mean cross-validation error of the three models.

1. Problem and data

The goal is to forecast total daily retail sales 14 days ahead. The data comes from the Kaggle competition *Store Sales - Time Series Forecasting*. Sales of all stores and product families are summed into one daily series from 2015-01-01 to 2017-08-15 (958 days). Earlier data was dropped because it is a ramp-up period. Days that are missing from the file (25 December, stores closed) were filled with 0 sales.

2. Approach

- **Seasonal Naive (baseline):** sales equal the same weekday of last week.
- **SARIMAX(1,1,1)(1,1,1,7) (statistical):** captures trend and weekly seasonality.
- **LightGBM (machine learning):** uses lag features (14, 21, 28, 35, 364 days), rolling mean and standard deviation of recent sales, calendar features (weekday, day, month, week of year, weekend), a national-holiday flag and a payday flag.

All lag and rolling features are at least 14 days old. A feature for any day in the forecast window therefore never uses unknown data, so there is no data leakage and no recursive forecasting is needed.

3. Validation

Models were compared with **time-based cross-validation** (`TimeSeriesSplit`, expanding window, 5 folds). In each fold the model is trained on the past and tested on the next 14 days. No random split is used. The same folds were used for all models, and metrics are MAE and RMSE averaged over the folds.

4. System architecture

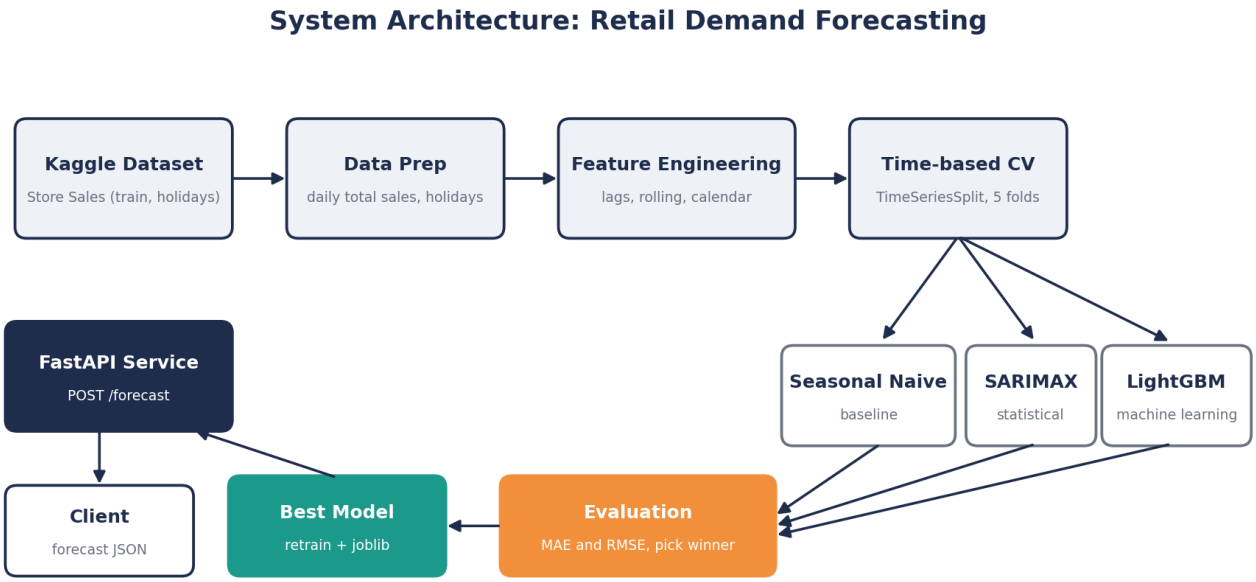


Figure 2. From raw data to the deployed API.

5. Results

Model	MAE	RMSE
LightGBM	**42,064**	**58,547**
SARIMAX	70,145	87,461
Seasonal Naive	98,696	128,217

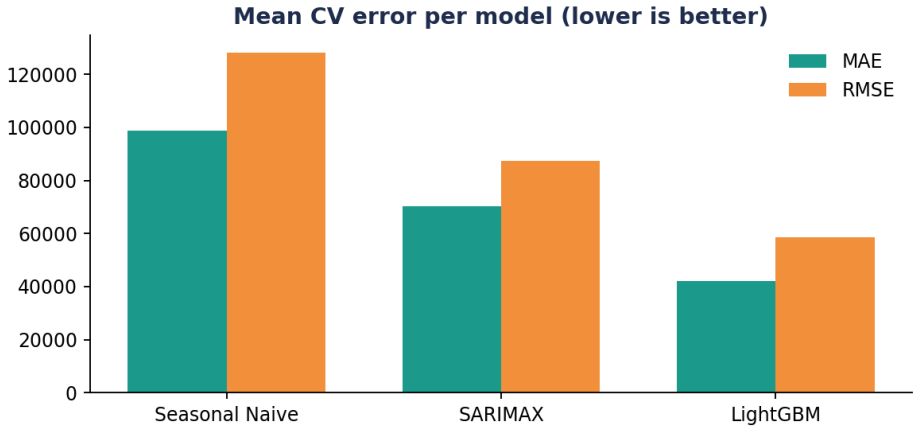


Figure 3. Mean CV error per model (lower is better).

LightGBM lowers MAE by 57% and RMSE by 54% compared with the Seasonal Naive baseline, and by 40% (MAE) and 33% (RMSE) compared with SARIMAX.

6. Why LightGBM won

- The baseline and SARIMAX model only the weekly pattern and the trend. They cannot react to one-off events such as national holidays or month-end and payday effects.
- LightGBM sees these effects directly through the holiday and payday flags, and its lag and rolling features capture both the recent sales level and the yearly pattern.
- SARIMAX clearly beats the baseline, so weekly seasonality is strong, but it is not enough on its own.
- The comparison is fair: all models used the same time-based folds and no leaked information.

LightGBM was therefore retrained on the full history and deployed.

7. Deployment

The best model is saved with joblib and served by a FastAPI application. Run `python -m src.train` to train and `uvicorn api.main:app --reload` to start the service.

Endpoint	Purpose
GET /health	Service status
GET /model-info	Model name, training date, CV scores
POST /forecast	Forecast for horizon of 1 to 14 days

Example request body: `{"horizon": 7}`. The response lists one forecast per date, starting the day after the last training date.

8. Limitations and next steps

- Only the total series is forecast, not individual stores or product families.
- Future promotions and oil prices are not used as features.
- Possible improvements: hyperparameter tuning, Prophet as another statistical model, per-store models.